

- a) The *type* tells Processing about the kind of value the variable will hold.
- b) The *name* is how you'll use the variable later in the code.
- c) The *value* is what the variable points to.

For example, `int a = 15`. This line of code creates an integer variable *a* that points to value 15.

Let's try to create a simple moving line animation with the help of variables.

```
int a = 0;

void setup() {
  size(640, 360);
  stroke(255);
  background(0);
}

void draw() {
  background(0);
  a = a + 2;
  line(a, 0, a, height/2);

  if(a > width) {
    a = 0;
  }
}
```

Let's try to decrypt the code:

1. `int a=0` denotes an integer variable with the name 'a' and value 0.
2. `Size (640, 360)` defines the window's size (*width*, *height*).

3. `Stroke(255)` sets the stroke colour to objects. Stroke property is applicable for lines and borders around objects.
4. I have mentioned the `background(0)` in the draw block; otherwise, each time the draw block iterates, 'a' increases by two, and therefore the line gets redrawn to two pixels. However, the lines drawn from the previous iterations remain in the view.

To get rid of the marks the line leaves behind, we simply remove the `background(0)` from the *setup* block and paste it to be the very first line of the *draw* block. When background code is in the *setup* block, it runs once at the beginning, making the background black. But this isn't enough; we need the background to be set to black in each loop to cover the lines drawn from the previous loops. The background, being in the first line, runs first and becomes the base layer. On each loop, the canvas is painted black and new elements get drawn on top of the black background. So, we have no marks.

Now, let's move on to the most important part of the code:

```
a = a + 2;
line(a, 0, a, height/2);

if(a > width) {
  a = 0;
}
```

- i. The first line of code is an increment operator, where the value of 'a' increases with every iteration.
- ii. The second line states that with every increment in the value of *a*, draw a line from the starting point (0, 0). Its height is half the length of the canvas. The increment happens in the *x* coordinate like 0, 2, 4,, while the *y* coordinate remains the same.
- iii. In the last part of the code, for the *if* block, we have defined a condition that if the length of the line is equal to the width of the canvas, then rather than disappearing, the line will move again from its starting point because we have changed the counter value of *a* to 0.

The idea behind animating things in Processing is manipulating the object's coordinates programmatically to change their location. **END** 🐧

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